THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Is 3465 divisible by 11? 1

(a) Yes **(b)** Only if the last digit is even
(c) No **(d)** Only if the number is a perfect square

2. Is 4518 divisible by 6? 1

(a) No **(b)** Only if the last digit is even
(c) Only if the number is a perfect square **(d)** Yes

3. For the number 2538, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6174 **(b)** 6264 **(c)** 6084 **(d)** 6183

4. The sum of the three consecutive numbers 12, 13 and 14 is: 1

(a) 36 **(b)** 42 **(c)** 39 **(d)** 37

5. The sum of the five consecutive numbers starting from 14 is: 1

(a) 85 **(b)** 80 **(c)** 70 **(d)** 75

6. Think of 7. Add 5, multiply by 2 and subtract 10. The result is: 1

(a) 15 **(b)** 9 **(c)** 14 **(d)** 19

7. The smallest digit that can replace * in 63*8 so that the number is divisible by 9 is: 1

(a) 1 **(b)** 2 **(c)** 8 **(d)** 5

8. The sum of three odd numbers is: 1

(a) a multiple of 3 **(b)** odd **(c)** always prime **(d)** even

9. Is 8239 divisible by 11? 1

(a) Only if the last digit is even **(b)** No
(c) Only if the number is a perfect square **(d)** Yes

10. Is 3724 divisible by 6? 1

(a) No **(b)** Only if the number is a perfect square
(c) Yes **(d)** Only if the last digit is even

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Use the divisibility rule for 11 to test 6578. 2

12. Apply one step of the Kaprekar routine to 5632. 2

13. Take 13. Add 6, multiply by 3, then subtract 18. What is the result? How is it related to 13? 2

14. Use the divisibility rule for 11 to test 4829. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Check whether 35640 is divisible by 8 and by 11. 3

16. Apply the Kaprekar routine to 3521 repeatedly until you reach 6174. Write each step. 3

17. Show that the sum of 47 and its reverse is divisible by 11. 3

OR

Check whether 73128 is divisible by 8 and by 11. 3

18. Apply the Kaprekar routine to 7840 repeatedly until you reach 6174. Write each step. 3

OR

Show that the sum of 63 and its reverse is divisible by 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Replace the stars so that the number 4*3* is divisible by both 5 and 9. Find all possibilities. 5

OR

Take 83 and its reverse 38. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

20. Replace the stars so that the number 6*5* is divisible by both 10 and 9. Find all possibilities. 5

OR

Take 61 and its reverse 16. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

END OF QUESTION PAPER — Total: 40 Marks

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